**ASSIGNMENT No: 02 (BI)**
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**Title:** RNA-Seq Data Analysis
**Code:**

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import ttest_ind

print("RNA-SEQ DATA ANALYSIS")
print("-" * 60)

data = {

"Gene":["Gene_A","Gene_B","Gene_C","Gene_D","Gene_E","Gene_F","Gene_G","Gene_H",
"Gene_I","Gene_J"],
   "Control_1":[100,200,50,300,500,120,80,400,60,150],
   "Control_2":[110,190,55,310,480,130,85,390,65,145],
   "Control_3":[105,210,52,295,510,125,82,410,58,155],
   "Treatment_1":[300,180,200,290,520,500,75,390,250,145],
   "Treatment_2":[320,170,210,305,530,480,80,400,270,150],
   "Treatment_3":[310,175,195,300,510,510,78,395,260,148]
}

df = pd.DataFrame(data)

print("\nRNA-Seq Count Data:")
print(df)

print("\n\nDATASET INFORMATION")
print("-" * 60)
print("Number of genes:",len(df))
print("Number of samples:",len(df.columns)-1)
print("\nMissing Values:")
print(df.isnull().sum())

print("\n\nQUALITY CONTROL")
print("-" * 60)

sample_columns = df.columns[1:]
library_sizes = df[sample_columns].sum()

print("\nTotal counts in each sample:")
print(library_sizes)
```

```
plt.figure(figsize=(10,5))
library_sizes.plot(kind="bar")
plt.title("RNA-Seq Library Size")
plt.xlabel("Samples")
plt.ylabel("Total Counts")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

print("\n\nNORMALIZATION")
print("-" * 60)

cpm = df.copy()

for sample in sample_columns:
    cpm[sample] = (df[sample]/df[sample].sum())*1000000

print("\nNormalized Expression Data (CPM):")
print(cpm.round(2))

print("\n\nDIFFERENTIAL GENE EXPRESSION")
print("-" * 60)

control_columns=["Control_1","Control_2","Control_3"]
treatment_columns=["Treatment_1","Treatment_2","Treatment_3"]

results=[]

for index,row in cpm.iterrows():
    gene=row["Gene"]
    control_values=row[control_columns].astype(float)
    treatment_values=row[treatment_columns].astype(float)
    control_mean=control_values.mean()
    treatment_mean=treatment_values.mean()
    log2fc=np.log2((treatment_mean+1)/(control_mean+1))
    statistic,p_value=ttest_ind(treatment_values,control_values,equal_var=False)
    results.append([gene,control_mean,treatment_mean,log2fc,p_value])

results_df=pd.DataFrame(
    results,
    columns=["Gene","Control_Mean","Treatment_Mean","Log2_Fold_Change","P_Value"]
)

results_df=results_df.sort_values("P_Value")
n=len(results_df)
```

```
results_df["Adjusted_P_Value"]=[
    min(p*n/rank,1)
    for rank,p in enumerate(results_df["P_Value"],start=1)
]

results_df=results_df.sort_values("Adjusted_P_Value")

results_df["Expression"]="Not Significant"

results_df.loc[
    (results_df["Adjusted_P_Value"]<0.05)&
    (results_df["Log2_Fold_Change"]>0),
    "Expression"
]="Upregulated"

results_df.loc[
    (results_df["Adjusted_P_Value"]<0.05)&
    (results_df["Log2_Fold_Change"]<0),
    "Expression"
]="Downregulated"

print("\nDifferential Expression Results:")
print(results_df.round(4))

plt.figure(figsize=(10,6))

plot_data=results_df.copy()
plot_data["-log10(p-value)"]=-np.log10(plot_data["P_Value"]+1e-10)

sns.scatterplot(
    data=plot_data,
    x="Log2_Fold_Change",
    y="-log10(p-value)",
    hue="Expression",
    s=100
)

plt.axvline(x=1,linestyle="--")
plt.axvline(x=-1,linestyle="--")
plt.axhline(y=-np.log10(0.05),linestyle="--")
plt.title("Differential Gene Expression - Volcano Plot")
plt.xlabel("Log2 Fold Change")
plt.ylabel("-Log10(P-value)")
plt.tight_layout()
plt.show()

expression_data=cpm.set_index("Gene")
```

```
plt.figure(figsize=(10,6))
sns.heatmap(np.log2(expression_data+1),annot=True,fmt=".1f")
plt.title("Gene Expression Heatmap")
plt.xlabel("Samples")
plt.ylabel("Genes")
plt.tight_layout()
plt.show()

print("\n\nFINAL RESULTS")
print("-" * 60)

print("\nUpregulated Genes:")
print(
   results_df[results_df["Expression"]=="Upregulated"]
   [["Gene","Log2_Fold_Change","Adjusted_P_Value"]]
   .round(4)
)

print("\nDownregulated Genes:")
print(
   results_df[results_df["Expression"]=="Downregulated"]
   [["Gene","Log2_Fold_Change","Adjusted_P_Value"]]
   .round(4)
)

print("\nRNA-Seq analysis completed successfully.")
print("Aditya Rade BEAD23167")
```

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**OUTPUT:**

Problems 10 | Debug Console | Output | Terminal | Ports | Jupyter

```
PS C:\Users\adity\OneDrive\Documents\Desktop\Sem 7 study materials\CL - 2> & C:\Users\adity\AppData\Local\Programs\Python
OneDrive/Documents/Desktop/Sem 7 study materials/CL - 2/Pracical-2-BI.py"
RNA-SEQ DATA ANALYSIS
------------------------------------------------------------

RNA-Seq Count Data:
     Gene  Control_1  Control_2  Control_3  Treatment_1  Treatment_2  Treatment_3
0  Gene_A        100        110        105          300          320          310
1  Gene_B        200        190        210          180          170          175
2  Gene_C         50         55         52          200          210          195
3  Gene_D        300        310        295          290          305          300
4  Gene_E        500        480        510          520          530          510
5  Gene_F        120        130        125          500          480          510
6  Gene_G         80         85         82           75           80           78
7  Gene_H        400        390        410          390          400          395
8  Gene_I         60         65         58          250          270          260
9  Gene_J        150        145        155          145          150          148
```

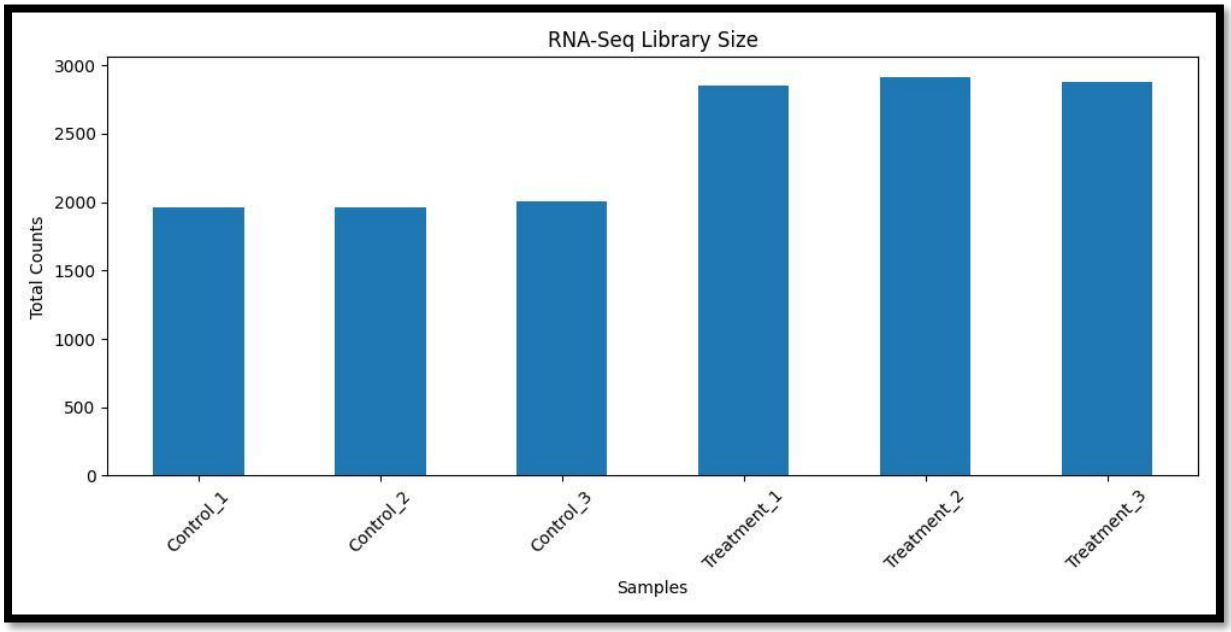
```
Problems 10  Debug Console  Output  Terminal  Ports  Jupyter

PS C:\Users\adity\OneDrive\Documents\Desktop\Sem 7 study materials\CL - 2> & C:\Users\adity\Ap
OneDrive/Documents/Desktop/Sem 7 study materials/CL - 2/Pracical-2-BI.py"
Number of genes: 10
Number of samples: 6

Missing Values:
Gene           0
Control_1      0
Control_2      0
Control_3      0
Treatment_1    0
Treatment_2    0
Treatment_3    0
dtype: int64


QUALITY CONTROL
------------------------------------------------------------

Total counts in each sample:
Control_1      1960
Control_2      1960
Control_3      2002
Treatment_1    2850
Treatment_2    2915
Treatment_3    2881
dtype: int64
```



```
ORF 4
Start Position: 36
End Position: 47
Length: 12
Sequence: ATGCGTAGCTAG

Analysis completed successfully.
Aditya Rade   BEAD23167
PS C:\Users\adity\OneDrive\Documents\Desktop\Sem 7 study materials\CL - 2> ^T^T^T
```

Problems 10 Debug Console Output Terminal Ports Jupyter

```
PS C:\Users\adity\OneDrive\Documents\Desktop\Sem 7 study materials\CL - 2> & C:\Users\adity\AppData\Loc
OneDrive/Documents/Desktop/Sem 7 study materials/CL - 2/Practical-1-BI.py"
Number of occurrences: 4

3. GC CONTENT
----------------------------------------
Number of G: 13
Number of C: 10
GC Content: 45.10%

4. CODING REGION IDENTIFICATION
----------------------------------------
Potential Coding Regions / ORFs:

ORF 1
Start Position: 1
End Position: 24
Length: 24
Sequence: ATGCGATCGATCGATGCTAGCTAG

ORF 2
Start Position: 14
End Position: 28
Length: 15
Sequence: ATGCTAGCTAGCTAA

ORF 3
Start Position: 31
End Position: 51
Length: 21
Sequence: ATGAAATGCGTAGCTAGCTAA
```